

JAVA SNACKS – HOT DOGS

(Saturday, April 4, 2026)

Instructions:

1. **Avoid using Artificial Intelligence(AI) tools.**
2. **Submission must occur EVERYDAY.**
3. **Each question is to be in a different file.**
4. **Each file is to be titled what the app is doing e.g QuestionOne can be named MileToKilometerConverter**
5. **For every day create a new folder in your local machine and save each file in the respective folders.**

Submission Guideline: It is safe to assume you have created a repo called JavaTasks(or its equivalent).

Inside our folder called easterEggs create the folder hotDogs and save all of today's tasks inside it

Push to git and paste the link to the repository on the provided Google Sheet document. (Put the link on Column D).

1. Write a program that reads in a number in miles and converts it to kilometers. (Note: 1 mile is 1.6 kilometers.)
2. Write a program that reads in the length of sides of an equilateral triangle and computes the area using the formula: $\text{area} = (\sqrt{3} / 4) \times (\text{length of sides})^2$.
3. Write a program that reads in the length of sides of an equilateral triangle and computes the volume of the triangular prism using: $\text{volume} = \text{area} \times \text{length}$.
4. Write a program that reads a number in meters, converts it to feet, and displays the result. (Note: 1 meter is 3.2786 feet.)
5. Write a program that converts square meters into ping. The program prompts the user to enter a number in square meters, converts it to ping, and displays the result. (Note: 1 square meter is 0.3025 ping.)
6. Write a program that reads the subtotal and the gratuity rate, then computes the gratuity and total. For example, if the user enters 10 for subtotal and 12% for gratuity rate, the program should display \$1.2 as gratuity and \$11.2 as total.

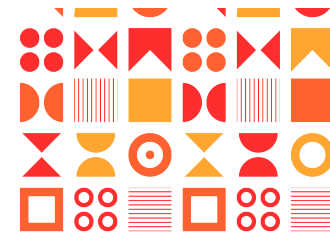
Website: www.semicolon.africa

Email: info@semicolon.africa

Tel: 0906 000 8609

Social: [@semicolonafrica](https://twitter.com/semicolonafrica)





7. Write a program that reads an integer between 0 and 1000 and adds all the digits in the integer. For example, if the integer is 932, the sum of all its digits is 14.
8. Write a program that prompts the user to enter the number of minutes (e.g., 1 billion), and displays the maximum number of years and remaining days for those minutes. (Assume a year has 365 days.).
9. Write a program that prompts the user to enter the initial velocity u in meters/second, time span t in seconds, and acceleration a in meters/second², then returns the distance covered using Newton's equation: $s = ut + \frac{1}{2}at^2$.
10. Write a program that calculates the energy needed to heat water from an initial temperature to a final temperature. Prompt the user to enter the amount of water in kilograms and the initial and final temperatures. Use the formula: $Q = M \times (\text{finalTemperature} - \text{initialTemperature}) \times 4184$.
11. Write a program that prompts the user to enter the number of years and displays the population after that many years. (Refer to Programming Exercise 1.11 for the population formula.)
12. Given an airplane's acceleration a and take-off speed v , write a program that prompts the user to enter v (m/s) and a (m/s²), then displays the minimum runway length needed using: $\text{length} = v^2 / (2a)$.
13. Write a program that prompts the user to enter a monthly saving amount and displays the account value after the sixth month, assuming an annual interest rate of 3.75% (monthly rate = 0.003125). Show the account value after each of the first six months.
14. Write a program that prompts the user to enter a weight in pounds and height in inches and displays the BMI. (Note: 1 pound = 0.45359237 kilograms, 1 inch = 0.0254 meters.)
15. Write a program that prompts the user to enter two points (x_1, y_1) and (x_2, y_2) and displays the distance between them using: $\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

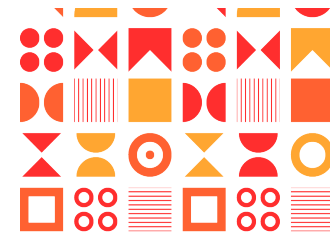
$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

16. Write a program that prompts the user to enter the side of a hexagon and displays its area using: $\text{Area} = (3\sqrt{3} / 2) \times s^2$.
17. Write a program that prompts the user to enter three points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) of a triangle and displays its area using:

$$s = (\text{side1} + \text{side2} + \text{side3})/2;$$

$$\text{area} = \sqrt{s(s - \text{side1})(s - \text{side2})(s - \text{side3})}$$

18. Write a program that reads the balance and the annual percentage interest rate, then displays the interest for the next month using: $\text{interest} = \text{balance} \times (\text{annualInterestRate} / 1200)$.



19. Write a program that reads in an investment amount, annual interest rate, and number of years, then displays the future investment value using:

$$\text{futureInvestmentValue} = \text{investmentAmount} \times (1 + \text{monthlyInterestRate})^{(\text{numberOfYears} \times 12)}$$

20. Write a program that prompts the user to enter the distance to drive, the fuel efficiency of the car in miles per gallon, and the price per gallon, then displays the cost of the trip.

21. Write a program that prompts the user to enter values for a , b , and c in the quadratic equation $ax^2 + bx + c = 0$ and displays the result. If the discriminant ($b^2 - 4ac$) is positive, display two roots. If it is zero, display one root. Otherwise, display: The equation has no real roots.

22. Write a program that prompts the user to enter three integers and displays them in non-decreasing order.

23. Write a program that prompts the user to enter a three-digit integer and determines whether it is a palindrome integer (reads the same from left to right as from right to left). A negative integer is treated the same as a positive integer.

24. Write a program that prompts the user to enter the exchange rate from U.S. dollars to Chinese RMB. Prompt the user to enter 0 to convert from dollars to RMB or 1 to convert from RMB to dollars. Prompt for the amount and display the converted result. Display an error for any code other than 0 or 1.

25. Write a program that prompts the user to enter the weight and price of two packages of rice and displays the one with the better price per unit weight.

26. Write a program that prompts the user to enter a year and determines whether it is a leap year. A year is a leap year if it is divisible by 4 but not 100, or if it is divisible by 400.

27. Write a program that prompts the user to enter an integer and determines whether it is divisible by 2, 3, 5, and/or 7. Print one line for each divisibility check.

28. Write a program that prompts the user to enter three sides of a triangle and determines whether the triangle is equilateral, isosceles, or scalene. Also determine whether the input forms a valid triangle (the sum of any two sides must be greater than the third).

29. Write a program that prompts the user to enter a score (0–100) and displays the corresponding letter grade: A (90–100), B (80–89), C (70–79), D (60–69), F (below 60).

30. Write a program that prompts the user to enter a number of seconds and displays it in hours, minutes, and remaining seconds (e.g., 3723 seconds = 1 hour, 2 minutes, 3 seconds).

Have fun guys!